Title page and declarations

Manuscript title

Calibration of Heart-Rate Reserve for Graded Cycle Ergometry: An Endpoint-Preserving Transfer Evaluated Against Linear Alternatives

Running title

Endpoint-preserving HRR calibration

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- Main tables: 4.
- References: 14.

Author contributions

BoTao Cai: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing of the original draft, review and editing, and project administration.

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Competing interests

The author declares no competing interests.

Ethics statement

No participants were recruited, contacted, or intervened upon, and no identifiable information was collected. The study analyzed public, deidentified datasets whose original records report ethics approval and consent. A separate institutional review or written exemption determination was not obtained for this secondary analysis.

Data availability

The development dataset is available from Zenodo at https://doi.org/10.5281/zenodo.10841412. The ACTES external dataset is available from PhysioNet at https://doi.org/10.13026/2qs3-kh43. Processed source data are supplied in the accompanying workbook.

Code availability

The complete reproducible analysis is available at https://github.com/xcai66/Calibration-of-Heart-Rate-Reserve-for-Graded-Cycle-Ergometry. The archived v2.0.0 release is available at https://doi.org/10.5281/zenodo.21860652; future versions are maintained under the Zenodo concept DOI https://doi.org/10.5281/zenodo.21689574.

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